A Spectral Remedy Frontier for Coordinated Polymer Repair

21 July 2026

Abstract

We convert the link-lift obstruction into an exact quantitative target for a coordinated multi-star repair.

Let λ bound external curvature energy relative to the repaired centered mode, let a be the contraction, and let $D_0 = m\eta^2$. Including fixed losses J, define

$$F(\eta, a, \lambda) = \frac{(1-a^2)D_0}{2t} - \frac{(1-a)^2\lambda D_0}{2t} - c(m-1)\log(1/a) - J.$$

Theorem 1. *A coordinated repair closes the channel whenever*

$$\lambda < \frac{(1-a^2) - \frac{2t}{D_0}(c(m-1)\log(1/a) + J + F_*)}{(1-a)^2}.$$

The inequality includes action, spillover, Jacobian, Haar, comparison, and target free energy.

Proof. Subtract the minimum external action increase from the selected-star action drop, then apply the change-of-variables determinant and the declared fixed losses. Solving $F > F_*$ for λ gives the criterion. □

Across the complete nonlinear split interval, the measured single-star spill coefficient fails. At the upper split it must be reduced to the calibrated admissible fraction recorded by the executable frontier.

Remark 1. *This is a noncircular design target. It does not construct the coordinated gauge-covariant repair.*